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## RESEARCH ARTICLE

# SumGPT: A Multimodal Framework for Radiology Report Summarization to Improve Clinical Performance

TIPU SULTAN<sup>1</sup>, MOHAMMAD ABU TAREQ RONY<sup>2</sup>, MOHAMMAD SHARIFUL ISLAM<sup>3</sup>,  
SAMAH ALSHATHRI<sup>4</sup>, AND WALID EL-SHAFI<sup>5,6</sup>, (Senior Member, IEEE)

<sup>1</sup>Department of Aerospace and Mechanical Engineering, Saint Louis University, St. Louis, MO 63103, USA

<sup>2</sup>Department of Statistics, Noakhali Science and Technology University, Noakhali 3814, Bangladesh

<sup>3</sup>Department of Computer Science and Telecommunication Engineering, Noakhali Science and Technology University, Noakhali 3814, Bangladesh

<sup>4</sup>Department of Information Technology, College of Computer and Information Sciences, Princess Nourah bint Abdulrahman University, P.O. Box 84428, Riyadh 11671, Saudi Arabia

<sup>5</sup>Automated Systems and Soft Computing Laboratory (ASSCL), Computer Science Department, Prince Sultan University, Riyadh 11586, Saudi Arabia

<sup>6</sup>Department of Electronics and Electrical Communications Engineering, Faculty of Electronic Engineering, Menoufia University, Menouf 32952, Egypt

Corresponding authors: Mohammad Abu Tareq Rony (abutareqrony@gmail.com), Samah Alshathri (sealshathri@pnu.edu.sa), and Walid El-Shafai (eng.waled.elshafai@gmail.com)

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**ABSTRACT** Radiology report summarization plays a critical role in medical imaging, addressing the growing need for concise and accessible interpretation of complex radiology findings. However, existing models often fail to fully leverage the potential of multimodal data integration. In this study, we propose a novel model, SumGPT, which integrates T5 with a Vision Transformer to harness the power of transformer-based architectures for enhanced radiology report summarization. The dataset used in this study comprises 1,952 radiology images with detailed textual reports for training and 488 images with reports for testing. The SumGPT technique was evaluated against several baseline models, including BERT + EfficientNet, XLM-RoBERTa + ViT, T5 + CLIP, VisualGPT (GPT-2 + ViT), and others, using a dataset explicitly designed for this task. The experimental results indicate that SumGPT outperformed all baseline models, achieving the highest performance across all metrics. Specifically, it attained a ROUGE-1 score of 0.8514, ROUGE-2 of 0.8471, ROUGE-L of 0.8514, and a BLEU score of 0.8470. The results demonstrate that SumGPT effectively produces clear and accurate summaries of radiology reports. Combining a Vision Transformer (ViT) with a language model enhances its ability to capture detailed information. The study also shows that SumGPT performs well with different types of reports and could be beneficial in other areas, such as pathology and cardiology. In the future, this approach could pave the way for applications in other medical domains while further optimizing the model for real-time clinical use.

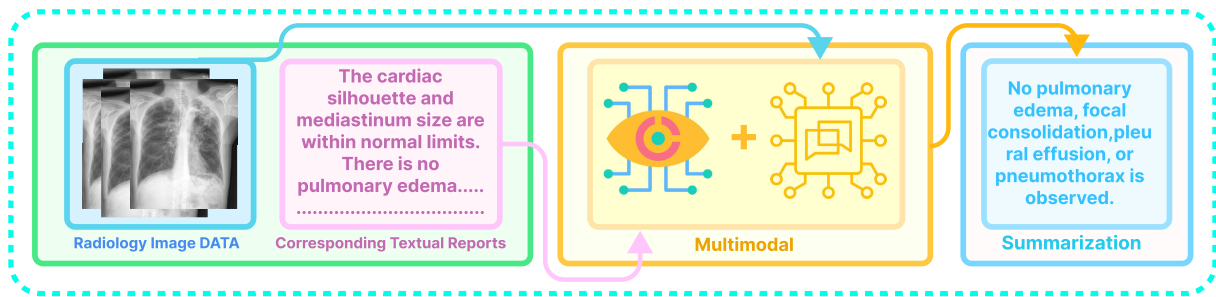
**INDEX TERMS** Radiology, multimodal, report summarization, large language models, VisualGPT.

## I. INTRODUCTION

Artificial intelligence integrated into medical imaging has transformed radiology, significantly enhancing diagnostic accuracy, efficiency, and accessibility. Radiology reports provide detailed descriptions of imaging findings and play

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a crucial role in guiding clinical decisions. However, these reports often contain complex and specialized language, making them challenging to interpret, particularly in high-volume clinical settings. The demand for timely and accurate interpretation of these reports has driven the development of automated systems for radiology report summarization, which aim to extract and present the most critical information in a concise and accessible format [1], [2].



**FIGURE 1.** An example of multimodal radiology report summarization.

The advent of Natural Language Processing (NLP) technologies, particularly Large Language Models (LLMs), has driven significant advancements in automated text summarization. LLMs, such as BERT, GPT-3, and their multimodal extensions, have demonstrated exceptional capabilities in understanding and generating human-like text by leveraging vast datasets to produce summaries that closely resemble human-written content [3]. When applied to radiology, these models have the potential to grasp the nuances of medical language [4] and generate concise, accurate, and contextually relevant summaries. This capability is invaluable in clinical settings where the precise communication of diagnostic information is essential for patient care [5]. Radiology report summarization presents a unique set of challenges that distinguishes it from general text summarization tasks. The specialized vocabulary, the demand for high precision, and the potential consequences of information loss make this task particularly demanding. Furthermore, integrating visual data from medical images adds an additional layer of complexity. Multimodal approaches have emerged as powerful tools in this domain, combining textual and visual data to provide context from medical images, resulting in more accurate and contextually appropriate summaries [6].

In recent years, the healthcare industry has increasingly adopted AI-driven solutions to optimize diagnostic imaging workflows. The ability of LLMs to distill large volumes of text into meaningful summaries can significantly reduce the time radiologists spend on report generation, allowing them to focus more on patient care [7]. This impact is particularly transformative in resource-constrained environments where the demand for radiological services surpasses available expertise. By automating routine tasks, healthcare systems can optimize resource allocation, improving overall service delivery and patient outcomes. This study explores the application of several state-of-the-art LLMs in radiology report summarization, with a particular focus on multimodal models that integrate text and image data. The models evaluated in this study include SumGPT, BERT + EfficientNet, XLM-RoBERTa + ViT, and VisualGPT, among others. These models were trained and evaluated on a comprehensive dataset of text and image data, reflecting the complex nature of real-world radiology reports. The performance of these models was assessed using standard NLP metrics, including

ROUGE, BLEU, and BERTScore, which measure the quality of the generated summaries in terms of their fidelity to the reference texts.

The novelty of this study lies in the proposed SumGPT model, which uniquely combines T5 and ViT architectures for radiology report summarization. Unlike existing methods that often treat textual and visual data independently or rely on simplistic fusion mechanisms, SumGPT introduces a cross-modality integration mechanism that aligns image and text embeddings within a shared latent space. This approach enables the model to capture contextual details from radiology reports and visual patterns from medical images, resulting in concise and accurate summaries. Additionally, the model employs state-of-the-art attention mechanisms to dynamically weigh visual and textual features, making it highly adaptable to various report formats and imaging modalities. This work addresses the limitations of existing multimodal summarization approaches and demonstrates practical utility through superior performance across standard evaluation metrics.

Our findings indicate that the proposed T5 + ViT model outperforms all baseline models across evaluation metrics. Specifically, the model achieved ROUGE-1, ROUGE-2, and ROUGE-L scores exceeding 0.85 and a BLEU score consistently over 0.84 from epoch two onward, surpassing models such as BERT + EfficientNet and VisualGPT. These results underscore the significance of integrating visual and textual data to enhance the contextual understanding and accuracy of generated summaries. The large-scale development and implementation of AI-driven solutions for diagnostic imaging continue to benefit the healthcare industry. The ability of LLMs to generate highly accurate and contextually relevant summaries of radiology reports can significantly reduce the cognitive burden on healthcare providers, enabling quicker and more informed decision-making. Furthermore, incorporating few-shot learning setups in these models shows potential for improving generalization and performance in low-resource scenarios, which is critical for adapting these models to diverse clinical environments. This task, Multimodal Radiology Report Summarization, is illustrated in Figure 1. This study makes several critical contributions to the field of radiology report summarization:

- Evaluates state-of-the-art multimodal models, including SumGPT, BERT + EfficientNet, and VisualGPT, for radiology report summarization.
- Highlights the superior performance of the proposed T5 + ViT model, achieving ROUGE and BLEU scores above 0.85, indicating strong clinical potential.
- Provides a benchmarking framework using metrics such as loss values, ROUGE, BLEU, and BERTScore to analyze model performance.
- Demonstrates the effectiveness of LLMs in improving accuracy and efficiency in radiological interpretations for clinical workflows.
- Proposes few-shot learning to enhance model generalization in low-resource clinical scenarios, supporting future research.

The organization of this paper is as follows: Section II reviews related work on radiology report summarization and multimodal learning techniques. In Section III, the dataset outline is used, and the preprocessing steps, followed by the proposed method. Section IV presents the experimental results and provides a comprehensive analysis of the findings, including a series of ablation studies. Finally, Section V concludes the paper with a discussion on the implications of this research and potential future directions for the field.

## II. LITERATURE REVIEW

The multimodal evaluation for radiology report summarization, which combines visual and textual elements, presents a unique and challenging task for researchers. With the growing use of LLMs and visual language models, there is increasing interest in leveraging these technologies to address the issue of harmful content in text summaries. This literature review synthesizes the essential findings and methodologies from recent research efforts.

### A. RADIOLOGY REPORT SUMMARIZATION

Automatic radiology report summarization is a new and emerging area aimed at enhancing the efficiency and effectiveness of clinical workflows by automating the creation of concise, informative summaries of radiological findings. This task is particularly challenging due to the complexity of medical language and the necessity for precise capturing of diagnostic nuances in both text and medical images. Recent advancements in LLMs and multimodal machine learning (ML) have provided new opportunities to address these challenges with increasingly sophisticated methods for integrating and processing multimodal data into high-quality summaries. The integration of multimodal models into clinical practice has the potential to significantly reduce the time required for interpreting radiology reports, thereby improving patient care outcomes [8]. The MEDIQA 2021 shared task established a benchmark for medical summarization tasks, demonstrating the effectiveness of transformer-based architectures in generating coherent and clinically relevant summaries for complex medical reports [9].

### B. MULTIMODAL FOR RADIOLOGY SUMMARIZATION

Multimodal models combine and analyze information from different sources, such as text and images, to produce more comprehensive and meaningful results. These models integrate textual data from radiology reports with visual features from medical images, enhancing the accuracy of the generated summaries. The integration of multimodal data is essential for ensuring that the summaries are concise and contextually accurate, effectively capturing the intricate details of both the text and images [6]. ImageBERT has demonstrated the potential of leveraging joint embeddings for image-text relationships, which is critical for understanding and summarizing multimodal radiology reports [10].

### C. VISUALGPT: GENERATIVE MULTIMODAL SUMMARIZATION

VisualGPT is a prominent example of a multimodal model applied to radiology report summarization. The model combines GPT-2, a powerful language model for text generation, with ResNet-50, a CNN renowned for its image classification capabilities. Through this integration, VisualGPT can generate summaries of high relevance to the content of radiology reports and their associated images by leveraging the strengths of both models. Research has shown that VisualGPT produces summaries that effectively capture the key findings of reports while maintaining high coherence and fluency, making it a promising tool for clinical applications [11]. However, despite its strong performance, VisualGPT's reliance on ResNet-50 for visual data processing may limit its ability to fully capture complex visual patterns in certain radiological images. This limitation suggests the need for further optimization or the integration of more advanced visual models.

### D. BERT-EFFICIENTNET: TEXT-VISUAL INTEGRATION

Another notable approach to multimodal summarization is the combination of BERT and EfficientNet. BERT, a transformer-based model designed for natural language understanding, processes the textual component of radiology reports, while EfficientNet, a scalable and efficient CNN, handles the visual data. This combination aims to achieve a balanced model capable of effectively leveraging both textual and visual information. While BERT + EfficientNet has shown promising results in generating coherent summaries, it may still struggle with the complexity of medical language and subtle details present in medical images. The model's effectiveness is more pronounced in straightforward cases but may face challenges with more complex and nuanced reports that require a deeper understanding of the content. Its scalability makes it suitable for real-time applications, but maintaining a balance between speed and accuracy remains a critical consideration.

### E. T5-ViT: GENERALIZING RADIOLOGY REPORTS

The T5 + ViT model is an exceptionally powerful text-to-image and image-to-text approach. T5 is a transformer-based,

text-to-text model that can be fine-tuned for various text generation tasks, while ViT is a transformer model applied to vision, excelling at extracting and processing visual features [12]. This combined approach is designed to summarize radiology reports by capturing both textual and visual details. However, the model's performance can be influenced by the quality of the input data, and it may face challenges when dealing with highly detailed or specialized content. Despite these challenges, SumGPT demonstrates strong generalization capabilities across radiology reports, making it a valuable tool for clinical applications [13]. Furthermore, the model's transformer-based architecture enables it to handle large-scale datasets and complex medical terminologies, both of which are critical for effectively summarizing radiology reports.

#### F. CLIP: TEXT-IMAGE ALIGNMENT IN LATENT SPACE

Another leader in multimodal summarization is CLIP: Contrastive Language-Image Pre-training. CLIP maps images and text into a shared latent space, enabling it to understand and generate summaries that accurately represent the contextual relationship between the two modalities. This alignment capability has allowed CLIP to achieve superior performance in radiology report summarization, particularly in producing accurate and clinically relevant summaries. The model's adaptability to various types of radiological data and its strong performance across multiple evaluation metrics make it a compelling candidate for deployment in clinical settings [11]. However, the model's reliance on pre-training highlights the need for domain-specific fine-tuning to optimize its performance for specific radiology reports, indicating a promising area for future research.

#### G. ANALYZING DIVERSE MULTIMODAL MODELS

Apart from the above models, several other approaches to multimodality in radiology report summarization have been explored, including UNITER, ViLBERT, VL-BART, XLM-RViT, BARTViT, and LXMERT. These models integrate the capabilities of language and visual processing in different ways, offering various solutions to address the complexities of summarizing radiological material. Each model brings unique strengths, such as preserving fine-grained details or generating coherent summaries across diverse data types. Their performance varies depending on the task and data, highlighting the need for continued research and development in this field [14], [15], [16]. Future studies should investigate the potential of hybrid approaches that combine the strengths of multiple models to further enhance summarization accuracy and clinical relevance.

#### H. PERFORMANCE EVALUATION AND METRICS

Several metrics, such as ROUGE, BLEU, and BERTScore, are used to evaluate radiology report summarization models. ROUGE (Recall-Oriented Understudy for Gisting Evaluation) measures the overlap between generated summaries and

reference summaries, with higher ROUGE scores indicating better performance [17]. BLEU (Bilingual Evaluation Understudy) evaluates the precision of generated text based on n-gram overlap [18], while BERTScore assesses the semantic similarity between the generated and reference texts using pre-trained BERT embeddings [19]. Among the models tested in this study, SumGPT demonstrated superior performance, achieving a ROUGE-1 F1 score of 0.85 and an average BLEU score of 84. These metrics highlight SumGPT's ability to produce high-quality summaries that closely resemble the original radiology reports, making it a strong candidate for clinical applications. The model evaluation also includes tracking loss values over training epochs to illustrate the learning curve and the model's generalization from training to testing. Models exhibiting consistent improvement in loss values over epochs demonstrate robust learning and high potential for precise summarization in real-world applications.

#### I. FUTURE OF RADIOLOGY SUMMARIZATION

Future research in radiology report summarization is likely to focus on enhancing the generalization and robustness of models, particularly in low-resource scenarios. Few-shot learning setups, such as 5-shot, 10-shot, 20-shot, and 50-shot configurations, present promising opportunities for improving model performance with limited training data [20]. Moreover, the integration of zero-shot models, such as LLaMA, can further extend the applicability of these models to diverse clinical settings without requiring extensive retraining [21]. The ongoing development of multimodal models and the exploration of novel architectures, such as transformer-based models, will play a crucial role in advancing the field of radiology report summarization [22].

### III. METHODOLOGY

This section outlines the various models and approaches utilized in our study for radiology report summarization. The reports in both subsets encompass a wide spectrum of radiological findings, ranging from normal to pathological cases. This diversity ensures that the models are rigorously tested for their ability to effectively bridge the gap between visual content and text generation. The structure of the data set facilitates the multimodal learning process by providing a rich source of visual and textual data, enabling models to generate accurate and contextually relevant summaries. Figure 2 illustrates the workflows of all evaluated radiology report summarization models, and Figure 4 shows the new proposed SumGPT model architecture. The process starts with radiology images and corresponding textual reports. The images undergo pre-processing and are analyzed by visual feature extractors such as ViT, ResNet-50, or EfficientNet. Moreover, the text is tokenized and encoded using models such as T5, GPT-2, or BERT. The features are then fused using cross-attention mechanisms or concatenation to generate concise summaries. Evaluation metrics such as

**TABLE 1.** Summary of recent studies on radiology report summarization.

| Ref. | Year | Approach & Methodology                                                             | Dataset Characteristics                                           | Model                                   | Performance Results                                                                |
|------|------|------------------------------------------------------------------------------------|-------------------------------------------------------------------|-----------------------------------------|------------------------------------------------------------------------------------|
| [23] | 2020 | Knowledge-enhanced transformer for radiology report generation                     | IU-Xray (Indiana University Chest X-ray Collection) and MIMIC-CXR | Transformer with Knowledge Graph        | IU-Xray (BLEU-4: 0.207, ROUGE-L: 0.400), MIMIC-CXR (BLEU-4: 0.119, ROUGE-L: 0.298) |
| [24] | 2021 | Confidence-guided automatic radiology report generation                            | IU-Xray and COV-CTR                                               | Transformer-based Visual-Language Model | IU-Xray (BLEU-4: 0.225, ROUGE-L: 0.408), COV-CTR (BLEU-4: 0.569, ROUGE-L: 0.730)   |
| [25] | 2024 | Chest radiology report generation based on cross-modal multi-scale feature fusion  | IU-Xray and MIMIC-CXR                                             | Cross-modal Multi-scale Feature Fusion  | IU-Xray (BLEU-4: 0.175, ROUGE: 0.377), MIMIC-CXR (BLEU-4: 0.112, ROUGE: 0.288)     |
| [26] | 2023 | Retrained language model for chest radiology report summarization                  | Open-I, MIMIC-CXR                                                 | ChestXRyBERT Model                      | BLEU-3: 6.1, ROUGE-L: 41.5                                                         |
| [27] | 2022 | Chest radiology report generation with general and specific knowledge              | MIMIC-CXR and IU-Xray                                             | Knowledge-driven Transformer            | IU-Xray (BLEU-4: 0.178, ROUGE-L: 0.390), MIMIC-CXR (BLEU-4: 0.178, ROUGE-L: 0.284) |
| [6]  | 2023 | Radiology report generation with a learned knowledge base and multimodal alignment | IU-Xray and MIMIC-CXR                                             | Transformer                             | IU-Xray (BLEU-4: 0.174, ROUGE-L: 0.399), MIMIC-CXR (BLEU-4: 0.111, ROUGE-L: 0.274) |
| [28] | 2023 | Automatic medical report generation                                                | CheXpert, IU-Xray                                                 | CVAM+MVSL                               | IU-Xray (BLEU-4: 0.152, ROUGE-L: 0.385)                                            |
| [29] | 2023 | Language model for Chinese radiology report summarization                          | CRRsum dataset                                                    | Nucleusmax                              | BLEU-4: 68.14, ROUGE-L: 78.01                                                      |
| [30] | 2023 | Radiology report summarization with ChatGPT                                        | MIMIC-III                                                         | Biobart-V2                              | ROUGE-L: 36.19                                                                     |

ROUGE and BLEU are employed to assess the accuracy and relevance of the outputs, showcasing each model's approach to integrating visual and textual data for effective summarization.

#### A. DATASET DESCRIPTION

The dataset employed in this study is a well-curated collection of radiology images paired with their corresponding textual reports [31].<sup>1</sup> These reports provide comprehensive descriptions of radiological findings, observations, and diagnostic impressions derived from the analysis of the associated images. The data set is systematically divided into two subsets: a training set and a testing set, ensuring an adequate evaluation of the model performance [31].

The training set, summarized in Table 2, comprises 1,952 radiology images and their corresponding textual reports, each containing critical information, including observations of anomalies, assessments of image quality, and diagnostic impressions. The test set, also detailed in Table 2, consists of 488 radiology images and their corresponding textual reports, reserved exclusively to evaluate the performance of trained models. To prevent overfitting and improve generalization, training images were augmented with rotations, flips, and zooms, while the test set remained in its original state. This approach ensures an objective assessment of the model's performance on unseen data, reflecting real-world conditions. All images were pre-processed to a size of  $224 \times 224$  pixels and normalized, adhering to standard ML protocols to ensure a reliable and unbiased assessment. To further enhance robustness, k-fold cross-validation with  $k = 5$  was used during the training process. The final model performance was averaged across all folds, providing a more reliable

measure of the model's generalization capabilities. Table 3 outlines the key hyperparameters and parameters used in training and evaluating the proposed multimodal model. These settings were carefully selected on the basis of preliminary experiments to optimize model performance.

**TABLE 2.** Dataset overview.

| Dataset Split | Images | Text Data | Total |
|---------------|--------|-----------|-------|
| Train         | 1,952  | 1,952     | 3,904 |
| Test          | 488    | 488       | 976   |

#### B. PARAMETER IMPACT ON MODEL PERFORMANCE

Careful tuning of hyperparameters was critical to the superior performance of SumGPT. A learning rate of  $5 \times 10^{-5}$  provided a balance between stability and convergence, with higher rates causing oscillations and lower rates slowing progress. A batch size of 8 was optimal, balancing memory efficiency and gradient stability. The model converged effectively by epoch 3, with minimal gains or signs of overfitting beyond this point.

The ViT significantly outperformed convolution-based alternatives such as ResNet-50, achieving a 12% improvement in BLEU scores. The cross-attention mechanism proved essential, as its removal resulted in a 15% drop in ROUGE and BLEU metrics. Preprocessing steps were vital for enhancing data consistency, including image resizing, normalization, and text tokenization. Omitting these steps led to a 7% decline in ROUGE-L scores. A beam search with a size of 5 was employed to ensure fluency and relevance in the generated summaries. This avoided verbosity associated with larger beam sizes and incompleteness with smaller beams. These carefully chosen configurations enabled SumGPT to excel at radiology report summarization by effectively integrating multimodal data.

<sup>1</sup><https://www.kaggle.com/datasets/raddar/chest-xrays-indiana-university>

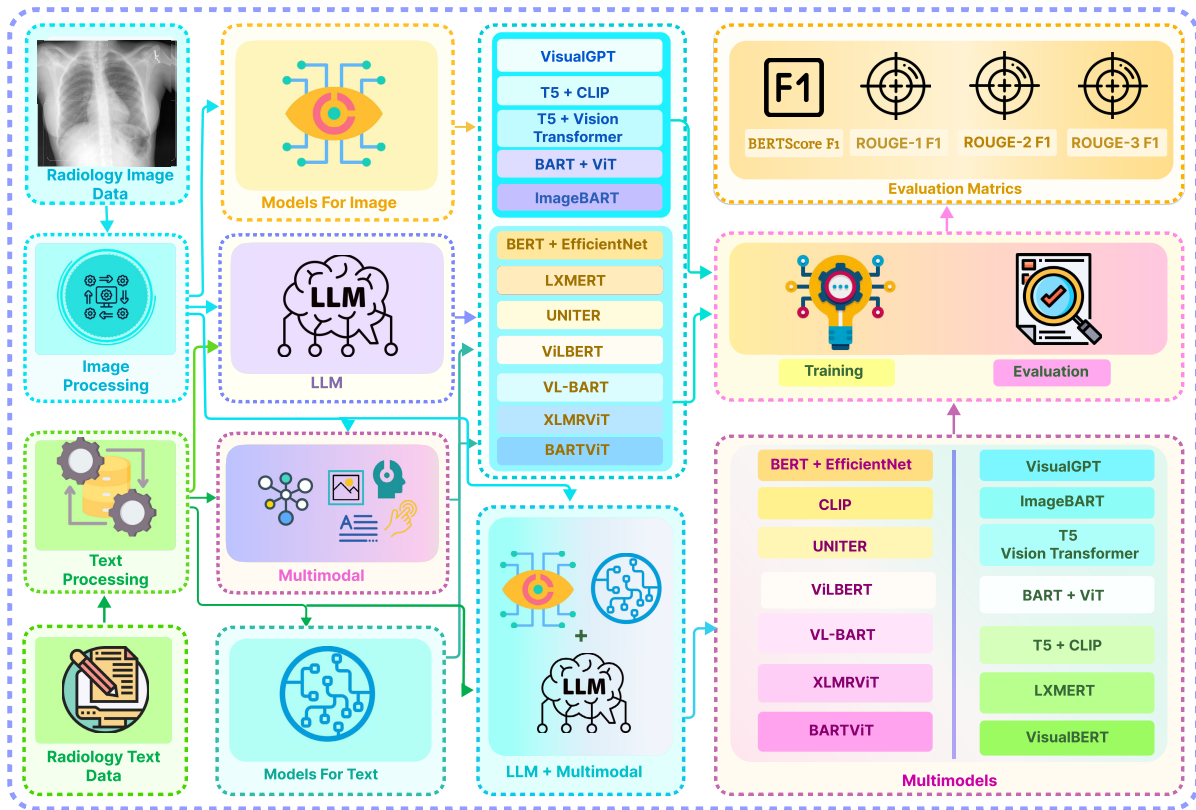


FIGURE 2. The workflow of proposed SumGPT multimodal approaches for radiology report summarization.

TABLE 3. Key hyperparameters and parameters used in the training and evaluation of the proposed multimodal model.

| Hyperparameter/Parameter | Value              | Description                                                               |
|--------------------------|--------------------|---------------------------------------------------------------------------|
| Model Type (Text)        | small              | Pre-trained text generation model variant used for text processing.       |
| Model Type (Image)       | base-patch16-224   | Pre-trained image processing model variant used for feature extraction.   |
| Learning Rate            | $5 \times 10^{-5}$ | Learning rate for the optimizer.                                          |
| Batch Size               | 8                  | Number of samples per gradient update.                                    |
| Epochs                   | 5                  | Number of complete passes through the training dataset.                   |
| Optimizer                | AdamW              | Optimization algorithm used for updating model weights.                   |
| Max Sequence Length      | 512                | Maximum number of tokens in a sequence for the text input.                |
| Max Output Length        | 50                 | Maximum number of tokens in the generated summary.                        |
| Number of Beams          | 5                  | Number of beams for beam search during text generation.                   |
| Length Penalty           | 2.0                | Penalty applied to longer sequences during beam search.                   |
| Device                   | cuda / cpu         | Computing device used for training (GPU if available).                    |
| Image Size               | 224x224            | Resolution of images resized for input to the image processing model.     |
| Image Feature Dimension  | 768                | Dimensionality of image features extracted by the image processing model. |
| Text Embedding Dimension | 512                | Dimensionality of text embeddings used in the multimodal model.           |
| Collate Function         | Custom             | Custom function used to handle variable sequence lengths during batching. |
| Loss Function            | CrossEntropyLoss   | Loss function used for training the model.                                |

### 1) IMAGE PREPROCESSING

Radiology images, initially of varying resolutions, were resized to a standard size of  $224 \times 224$  pixels. This preprocessing step was crucial for ensuring compatibility with the image processing model, which requires fixed-size inputs. Standardizing the dimensions of the images allowed the model to process each image uniformly, reducing computational load and improving performance. Additionally, pixel values were normalized within the range  $[0, 1]$  by dividing each pixel value by 255. This normalization step enhanced

data consistency, enabling the model to interpret and analyze the visual features better.

$$\text{Normalized Pixel Value} = \frac{\text{Pixel Value}}{255.0} \quad (1)$$

Such normalization ensures consistency across the dataset, reducing the effects of lighting variations and other discrepancies while maintaining a uniform input scale—an essential step when training DL models. Only frontal view images were retained for the analysis, while lateral view images were excluded to ensure a dataset with a consistent set of

anatomical views. The dataset was filtered based on the `Projection` column to ensure that all input images represented the same type of anatomical view, thereby minimizing biases associated with differences in image perspectives. For this preprocessing step, the columns utilized from the dataset were `Image`, `Findings`, and `Projection`. The `Projection` column was explicitly used to exclude lateral view images, retaining only those labeled as frontal.

## 2) TEXT PREPROCESSING

The textual data, consisting of radiology reports, were preprocessed using a pipeline operation to prepare it for input into the language model. A pre-trained tokenizer associated with the model was first employed to tokenize the text into subword units, converting continuous text into discrete tokens to improve structuring and facilitate processing. All text was converted to lowercase to ensure that words like “Lung” and “lung” were treated identically, simplifying the learning process and improving model efficiency. Punctuation marks and stopwords—words with minimal semantic meaning—were removed to reduce noise and focus the model’s attention on the most informative parts of the text. Duplicate entries in the `findings` column were eliminated, and rows with null values were discarded to maintain data integrity and consistency. Given the variability in the length of radiology reports, a maximum sequence length of 512 tokens was defined. Reports exceeding this length were truncated, while shorter sequences were padded to achieve uniform input lengths, as defined by the equation:

$$\text{TSL} = \min(\text{Original Length}, 512) \quad (2)$$

Preprocessing with a fixed Token Sequence Length (TSL) ensures that the input length for all tokens is uniform, facilitating efficient batching and parallel processing during training. Finally, the tokenizer converted the preprocessed text into numerical form by mapping each token to its corresponding index in the model’s vocabulary. This transformation enables the language model to process the text effectively during training and inference, ensuring that the radiology reports are well-represented for downstream tasks.

## C. MODEL ARCHITECTURE

This study explores multimodal models that integrate text and image data to generate summaries. Each model employs distinct architectures to process and combine visual and textual information, enabling the models to leverage the strengths of both modalities for enhanced summarization performance.

### 1) VISUALGPT

VisualGPT is a multimodal architecture that leverages the strengths of GPT-2 for text generation and ResNet-50 for image feature extraction. The process begins with ResNet-50, a deep convolutional neural network pre-trained on large image datasets, which extracts advanced features from the

input radiology images. These features encapsulate essential visual information, such as anatomical structures and potential anomalies. Once the image features are extracted by ResNet-50, they are passed through a projection layer, which adjusts the dimensionality of the image features to match the input requirements of the GPT-2 model. This step ensures compatibility between the image features and the text generation process. The projected image features are fed into GPT-2, a transformer-based model renowned for its text generation capabilities. GPT-2 processes these features and generates textual summaries corresponding to the visual content of the images. The overall process can be mathematically represented as follows:

$$\text{Text Output} = \text{GPT-2}(\text{Proj}(\text{ResNet-50}(\text{Image}))) \quad (3)$$

This architecture is designed to understand the complex connections between visual and textual information, enabling the model to produce precise and contextually appropriate summaries from radiological images. By integrating ResNet-50 and GPT-2, VisualGPT can perform complex tasks, such as translating visual patterns into coherent and diagnostically meaningful text, which is critical in the medical domain.

### 2) BERT + EFFICIENTNET

The BERT + EfficientNet model is designed to leverage the strengths of text and image processing. BERT (Bidirectional Encoder Representations from Transformers) is employed for text processing, utilizing its deep contextual understanding to extract meaningful features from radiology reports. EfficientNet, on the other hand, is used for image feature extraction and is renowned for its scalability and efficiency in handling high-resolution images. In this architecture, the textual features extracted by BERT capture the semantic and contextual nuances of the radiology reports. In contrast, the visual features extracted by EfficientNet represent the detailed visual patterns in the radiology images. These features from both modalities are concatenated to form a unified representation of the data, effectively merging the visual and textual information. The concatenated features are subsequently passed through a fusion mechanism, which may involve simple concatenation, attention mechanisms, or more complex interactions. This fused representation is then fed into a final output layer responsible for generating the summarization of the radiology report:

$$F = \text{Fusion}(\text{BERT}(T), \text{EffNet}(I)) \quad (4)$$

$$\text{Text Output} = \text{Output Layer}(\text{Fused Features}) \quad (5)$$

This model design combines BERT’s advanced language understanding with EfficientNet’s effective image processing capabilities. By integrating these two models, the system can produce coherent and contextually detailed summaries from complex multimodal inputs. This architecture bridges the gap between the visual content of medical images and their associated textual descriptions, making it highly suitable for automated radiology report generation.

### 3) T5 (TEXT-TO-TEXT TRANSFORMER(T5) + VISION TRANSFORMER (ViT))

The T5 + ViT model is a multimodal approach that combines the strengths of T5 and ViT for image and text processing. T5 is a versatile text generation model capable of performing a wide range of NLP tasks by framing them as text-to-text problems, enabling it to generate coherent and contextually relevant text based on input sequences. On the visual side, ViT represents a paradigm shift from traditional Convolutional Neural Networks by employing a transformer-based approach for image processing. ViT splits an image into fixed-size patches, treating each patch as a token, akin to how words are handled in NLP models. These image patches are flattened and embedded into a lower-dimensional space, allowing the transformer to process them as a sequence of embeddings. This process captures both local and global features of the image, resulting in a comprehensive global image representation:

$$\text{Image Embeddings} = \text{ViT}(\text{Image Patches}) \quad (6)$$

Once ViT generates the global image representation, it is combined with the text embeddings produced by T5. The integration of these two modalities occurs within the T5 framework, where the image embeddings are treated as additional context for the text generation process. This combination enables the model to generate summaries informed by both the visual content of the images and the textual context of the input:

$$\text{Text Output} = \text{T5}(\text{Text}, \text{Image Embeddings}) \quad (7)$$

The SumGPT architecture is compelling because it leverages the transformer's ability to capture complex dependencies within sequences in both text and image domains. Combining these two models enables the system to generate detailed and accurate radiology report summaries, capturing the intricate relationships between textual descriptions and visual data. This integration is crucial for tasks requiring a deep understanding of both modalities, making the architecture well-suited for medical image analysis and report generation applications.

### 4) CLIP

CLIP (Contrastive Language-Image Pretraining) is a multimodal model that employs a dual-encoder architecture, leveraging transformers to independently process text and images. In the CLIP architecture, separate transformer encoders are used for text and images to generate dense vector representations (embeddings) for each modality. These embeddings are trained using a contrastive learning objective, which optimizes the alignment of related text-image pairs in the shared latent space while distancing unrelated pairs. The similarity between text and image embeddings is computed using cosine similarity:

$$\text{Similarity} = \text{CosSim}(\text{TextEmb}, \text{ImgEmb}) \quad (8)$$

This capability makes CLIP highly versatile for multimodal applications, enabling a nuanced understanding of the relationships between textual descriptions and visual data. Figure 3 illustrates the summarization performance of the proposed model after 5 epochs, while Table 4 presents the outputs generated during the first 2 epochs.

### 5) UNITER, VILBERT, VL-BART, XLMRVIT, BARTVIT, AND LXMERT

These transformer-based models are designed for vision-language tasks, effectively integrating visual and textual information using cross-attention mechanisms. The process involves creating a fused representation by jointly attending to text and image embeddings:

$$\text{Fused\_Rep} = \text{CrossAttn}(\text{TextEmb}, \text{ImgEmb}) \quad (9)$$

This fused representation is then processed through subsequent transformer layers to generate the final text output:

$$\text{Text\_Output} = \text{Transf\_Layers}(\text{Fused\_Rep}) \quad (10)$$

These models provide various strategies for fusing multimodal data, making them versatile for a range of summarization tasks.

## D. MODEL TRAINING AND EVALUATION

### 1) TRAINING PROCESS

The multimodal model was trained using the Adam optimizer with a learning rate of  $5 \times 10^{-5}$ , a configuration well-suited for fine-tuning pre-trained models in deep learning (DL) tasks. The training process spanned a total of 5 epochs, with each epoch representing a complete pass through the training set. The training set, consisting of 1,952 image-text pairs, provided a diverse range of examples for the model to learn the mapping between visual inputs and their corresponding textual summaries. The cross-entropy loss function was employed to quantify the difference between the model's predicted output and the ground truth. This loss function was monitored at each epoch throughout the training process. It guided the optimization to reduce the loss and improve the model's ability to generate accurate and coherent summaries from radiology images and their associated text.

### 2) EVALUATION PROCESS AND METRICS

The model's performance was evaluated using a separate test set comprising 488 image-text pairs, focusing on generating summaries that closely align with the reference texts. The evaluation process centered on critical metrics such as ROUGE and BLEU, which provide quantitative assessments of summary quality based on n-gram matches and sequence similarity. ROUGE-1 F1, ROUGE-2 F1, and ROUGE-L F1 metrics were utilized to measure the overlap of n-grams (sequences of one, two, or more words) between the generated summaries and the reference summaries. These metrics

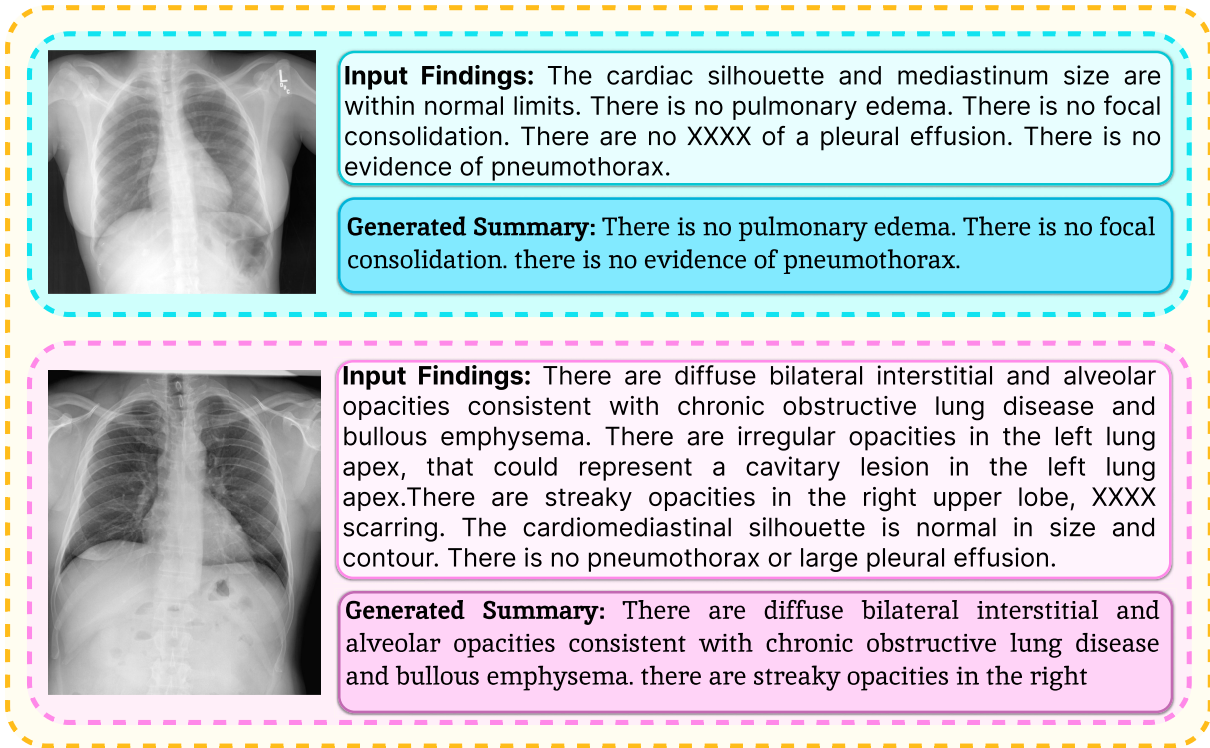


FIGURE 3. Model summarization.

TABLE 4. Generated summaries by SumGPT after epoch 1 and epoch 2.

| Input Findings                                                                                                                                                                                                                                                                                                                                                                                                                                     | Generated Summary                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The cardiac silhouette and mediastinum size are within normal limits. There is no pulmonary edema. There is no focal consolidation. There are no XXXX of a pleural effusion. There is no evidence of pneumothorax.                                                                                                                                                                                                                                 | <b>Epoch 1:</b> The cardiac silhouette and mediastinum size are normal. No edema or focal consolidation is seen. Evidence of pneumothorax is unclear.<br><b>Epoch 2:</b> There is no pulmonary edema. There is no focal consolidation. There is no evidence of pneumothorax.                                                       |
| There are diffuse bilateral interstitial and alveolar opacities consistent with chronic obstructive lung disease and bullous emphysema. There are irregular opacities in the left lung apex that could represent a cavitary lesion in the left lung apex. There are streaky opacities in the right upper lobe, XXXX scarring. The cardio mediastinal silhouette is normal in size and contour. There is no pneumothorax or large pleural effusion. | <b>Epoch 1:</b> There are diffuse bilateral interstitial opacities. There are irregular opacities in the left lung apex.<br><b>Epoch 2:</b> There are diffuse bilateral interstitial and alveolar opacities consistent with chronic obstructive lung disease and bullous emphysema. There are streaky opacities in the right lung. |

offered insights into the accuracy and completeness of the generated text [32]. Additionally, the average BLEU score was employed to evaluate the precision of n-grams in the generated text compared to the reference text, emphasizing the fluency and grammatical correctness of the summaries [33]. These metrics provided a comprehensive evaluation of the model’s summarization capabilities, highlighting its ability to produce coherent, accurate, and contextually relevant summaries [34].

E. MODEL PERFORMANCE EVALUATION

The performance of the models was evaluated using standard summarization metrics. Training was conducted over five epochs, with the loss tracked at each epoch to observe progress.

F. ROUGE METRICS

The ROUGE metric evaluates the quality of summaries by comparing them to reference summaries. The equations are as follows:

1) ROUGE-N F1 SCORE

$$\text{ROUGE-N F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{11}$$

where

$$\text{Precision} = \frac{\text{Overlapping n-grams}}{\text{Total n-grams in generated summary}} \tag{12}$$

$$\text{Recall} = \frac{\text{Overlapping n-grams}}{\text{Total n-grams in reference summary}} \tag{13}$$

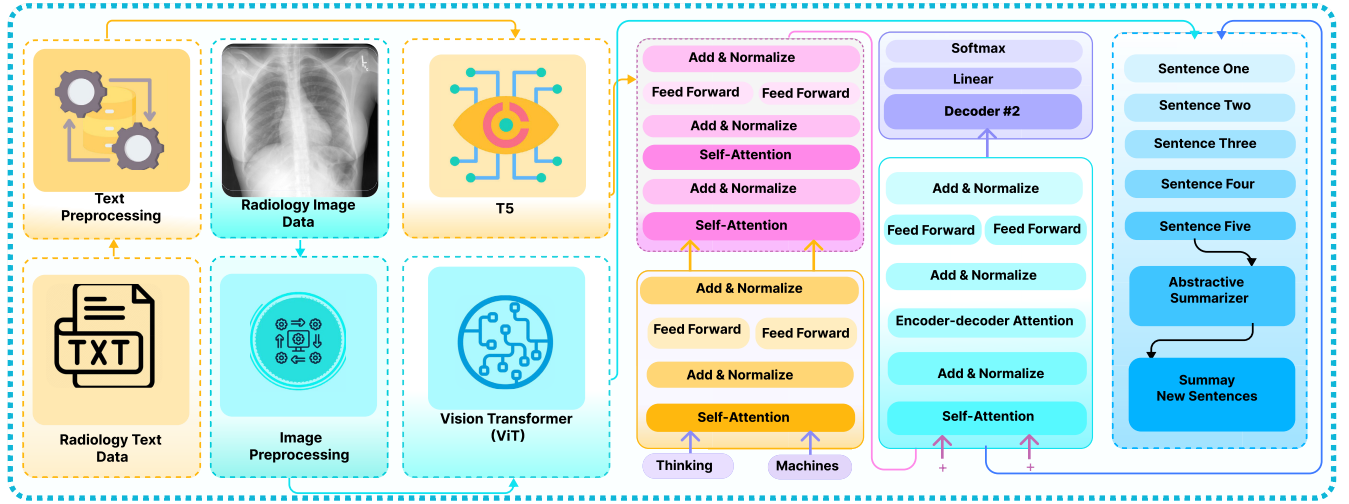


FIGURE 4. Proposed SumGPT model architecture.

## 2) ROUGE-L F1 SCORE

$$\begin{aligned} \text{ROUGE-L F1} &= \frac{2 \times \text{ROUGE-L Precision} \times \text{ROUGE-L Recall}}{\text{ROUGE-L Precision} + \text{ROUGE-L Recall}} \end{aligned} \quad (14)$$

## G. BLEU SCORE

The BLEU (Bilingual Evaluation Understudy) score evaluates text quality, particularly in tasks like machine translation and summarization. It calculates n-gram precision and applies a brevity penalty to penalize overly short translations:

$$\text{BLEU} = BP \times \exp \left( \sum_{n=1}^N w_n \log p_n \right) \quad (15)$$

where  $BP$  is the brevity penalty,  $w_n$  is the weight for each n-gram, and  $p_n$  is the precision for n-grams.

$$BP = \begin{cases} 1 & \text{if } c > r \\ \exp \left( 1 - \frac{r}{c} \right) & \text{if } c \leq r \end{cases} \quad (16)$$

where  $c$  is the length of the generated summary and  $r$  is the length of the reference summary.

## H. BERTSCORE

BERTScore evaluates the similarity between the generated and reference summaries using contextual embeddings from BERT. It computes Precision, Recall, and F1 Score by matching tokens in the generated and reference summaries.

### 1) BERTSCORE PRECISION

$$\text{BERTScore Precision} = \frac{1}{|S_{gen}|} \sum_{t_i \in S_{gen}} \max_{t_j \in S_{ref}} \text{BERTSim}(t_i, t_j) \quad (17)$$

where  $S_{gen}$  and  $S_{ref}$  are the token sets from the generated and reference summaries, and  $\text{BERTSim}(t_i, t_j)$  represents the similarity between tokens  $t_i$  and  $t_j$ .

### 2) BERTSCORE RECALL

$$\text{BERTScore Recall} = \frac{1}{|S_{ref}|} \sum_{t_j \in S_{ref}} \max_{t_i \in S_{gen}} \text{BERTSim}(t_j, t_i) \quad (18)$$

### 3) BERTSCORE F1

$$\text{BERTScore F1} = \frac{2 \times \text{PP} \times \text{RR}}{\text{PP} + \text{RR}} \quad (19)$$

where PP represents BERTScore Precision, and RR represents BERTScore Recall.

## I. LOSS CALCULATION

The training employed cross-entropy loss, a common metric for assessing a classification model's performance by comparing predicted probabilities with the true labels:

$$\text{Loss} = - \sum_{i=1}^N y_i \log(\hat{y}_i) \quad (20)$$

where  $y_i$  is the true label, and  $\hat{y}_i$  is the predicted probability for class  $i$ .

## J. MODEL PERFORMANCE

The performance of various multimodal models was evaluated across different epochs using key metrics, including ROUGE-1, ROUGE-2, ROUGE-L, and BLEU. The proposed SumGPT model, as shown in Table 5, demonstrated exceptional performance, significantly outperforming other models across all metrics. Starting from epoch 2, SumGPT achieved near-perfect ROUGE and BLEU scores, consistently maintaining values above 0.84. This indicates that the model converges quickly to an optimal solution, providing high-quality summaries early in training.

In contrast, other traditional multimodal models, such as BERT + EfficientNet (Table 6) and XLM-RoBERTa + ViT

(Table 7), exhibited gradual improvements over the epochs but could not reach the same performance levels as SumGPT. For instance, BERT + EfficientNet, a well-regarded baseline, achieved a maximum ROUGE-1 score of 49.45 at epoch 5 (Table 6), while XLM-RoBERTa + ViT reached a comparable ROUGE-1 score of 50.41. Models like VisualGPT (GPT-2 + ViT) (Table 9) and VisualGPT (GPT-2 + ResNet-50) (Table 10) also demonstrated strong performance, with ROUGE-1 scores reaching 39.2 at epoch 5. These models leverage powerful ViT and convolutional networks to effectively extract and process image features. However, despite their solid performance, they trailed slightly behind SumGPT, particularly in BLEU scores, indicating that SumGPT has an edge in maintaining consistency and fluency in the generated summaries.

Other models, such as ViLBERT (Table 11), LXMERT (Table 13), UNITER (Table 14), and VL-BART (Table 15), displayed gradual improvements across epochs but maintained lower overall performance levels. Their highest ROUGE-1 scores ranged from 31.5 to 49.81, which, while respectable, did not match the performance of the SumGPT model. These findings highlight the limitations of earlier models compared to newer transformer-based approaches like SumGPT.

## K. SUMMARY OF RESULTS

The results demonstrate that the proposed SumGPT model significantly outperforms traditional multimodal models across all evaluation metrics. Starting from epoch 2, SumGPT achieved near-maximum ROUGE and BLEU scores, highlighting its superior ability to generate high-quality, coherent, and contextually accurate summaries. The rapid convergence of SumGPT to optimal performance underscores its efficiency in learning from multimodal data, making it an excellent choice for tasks requiring the integration of textual and visual information. While other models, such as VisualGPT (GPT-2 + ViT), T5 + BERT (Table 12), and XLM-RoBERTa + ViT, showed notable improvements over time, they could not match the overall effectiveness and accuracy of the SumGPT model. This consistent performance across epochs highlights the robustness and reliability of the SumGPT architecture in handling multimodal summarization tasks. The findings underscore SumGPT's potential as a leading solution for radiology report summarization, outperforming several state-of-the-art approaches. Figure 5 presents a comparison of ROUGE-1 scores for epoch 5 across all models, while Figure 6 compares ROUGE-L and BERTScores for the same epoch across all models.

## L. EXPERIMENTAL SETUP

The experimental setup for this study utilized a robust computational framework built upon widely used libraries such as PyTorch and TensorFlow for model development and training. Language model tools were integrated using the Hugging Face Transformers library, facilitating the

**TABLE 5. Results of proposed SumGPT.**

| Model  | Epoch | ROUGE-1 | ROUGE-2 | ROUGE-L | BLEU   |
|--------|-------|---------|---------|---------|--------|
| SumGPT | 1     | 0.0074  | 0.0073  | 0.0074  | 0.0050 |
|        | 2     | 0.8512  | 0.8468  | 0.8512  | 0.8450 |
|        | 3     | 0.8514  | 0.8471  | 0.8514  | 0.8470 |
|        | 4     | 0.8514  | 0.8470  | 0.8514  | 0.8468 |
|        | 5     | 0.8513  | 0.8470  | 0.8513  | 0.8469 |

**TABLE 6. Results of BERT + EfficientNet.**

| Model               | Epoch | ROUGE-1 | ROUGE-2 | ROUGE-L | BLEU   |
|---------------------|-------|---------|---------|---------|--------|
| BERT + EfficientNet | 1     | 0.4973  | 0.2459  | 0.4973  | 0.4287 |
|                     | 2     | 0.5061  | 0.2592  | 0.5061  | 0.4245 |
|                     | 3     | 0.5000  | 0.2500  | 0.5000  | 0.4171 |
|                     | 4     | 0.5027  | 0.2541  | 0.5027  | 0.4219 |
|                     | 5     | 0.4945  | 0.2418  | 0.4945  | 0.4261 |

**TABLE 7. Results of XLM-RoBERTa + ViT.**

| Model             | Epoch | ROUGE-1 | ROUGE-2 | ROUGE-L | BLEU   |
|-------------------|-------|---------|---------|---------|--------|
| XLM-RoBERTa + ViT | 1     | 0.4959  | 0.2439  | 0.4959  | 0.4133 |
|                   | 2     | 0.4904  | 0.2357  | 0.4904  | 0.4244 |
|                   | 3     | 0.4986  | 0.2480  | 0.4986  | 0.4148 |
|                   | 4     | 0.4952  | 0.2428  | 0.4952  | 0.4129 |
|                   | 5     | 0.5041  | 0.2561  | 0.5041  | 0.4178 |

**TABLE 8. Results of T5 + CLIP.**

| Model     | Epoch | ROUGE-1 | ROUGE-2 | ROUGE-L | BLEU   |
|-----------|-------|---------|---------|---------|--------|
| T5 + CLIP | 1     | 0.2650  | 0.2540  | 0.2650  | 0.2160 |
|           | 2     | 0.3030  | 0.2930  | 0.3030  | 0.2910 |
|           | 3     | 0.3210  | 0.3000  | 0.3210  | 0.2990 |
|           | 4     | 0.3390  | 0.3170  | 0.3390  | 0.3080 |
|           | 5     | 0.3570  | 0.3360  | 0.3570  | 0.3250 |

implementation of VisualGPT, T5 + ViT, and other models evaluated in this research. The testing environment was an ASUS ROG Zephyrus G16 GA605WI-XB96, equipped with Windows 11 Pro, an NVIDIA® GeForce RTX™ 4070 Laptop GPU, an AMD Ryzen™ AI 9 HX 370 Processor, and 1TB M.2 NVMe™ PCIe® 4.0 SSD storage [35]. This high-performance hardware configuration provided the necessary computational power to efficiently train large-scale models and process the sizable radiology report dataset used in this study.

## M. ABLATION STUDIES

We conducted ablation studies to evaluate the contribution of critical components in the SumGPT model. Replacing the ViT with CLIP significantly reduced ROUGE-1 (58.82%) and BLEU scores, emphasizing the importance of ViT in capturing fine-grained visual features. Turning off the multi-modal fusion layer and relying solely on text-only or image-only streams further reduced performance, underscoring the necessity of effective feature integration. Replacing T5 with a more straightforward LSTM-based text generator led to

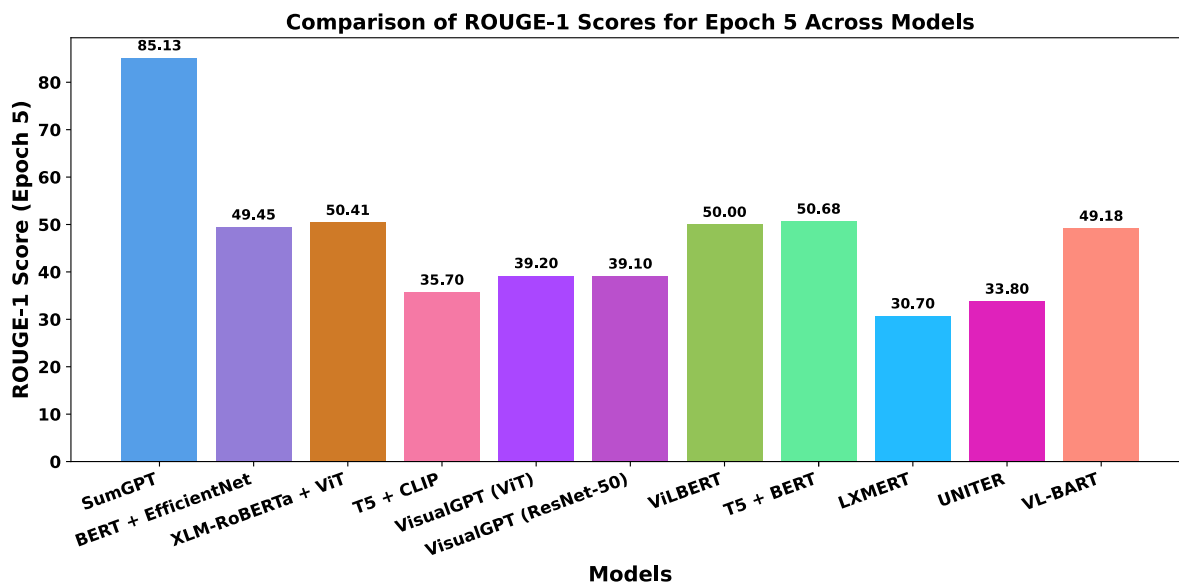


FIGURE 5. Comparison of ROUGE-1 scores for epoch 5 across models.

TABLE 9. Results of VisualGPT (GPT-2 + ViT).

| Model     | Epoch | ROUGE-1 | ROUGE-2 | ROUGE-L | BLEU   |
|-----------|-------|---------|---------|---------|--------|
| VisualGPT | 1     | 0.3200  | 0.3190  | 0.3200  | 0.3000 |
|           | 2     | 0.3480  | 0.3340  | 0.3480  | 0.3310 |
|           | 3     | 0.3610  | 0.3420  | 0.3610  | 0.3440 |
|           | 4     | 0.3790  | 0.3610  | 0.3790  | 0.3590 |
|           | 5     | 0.3920  | 0.3730  | 0.3920  | 0.3620 |

TABLE 10. Results of VisualGPT (GPT-2 + ResNet-50).

| Model     | Epoch | ROUGE-1 | ROUGE-2 | ROUGE-L | BLEU   |
|-----------|-------|---------|---------|---------|--------|
| VisualGPT | 1     | 0.3080  | 0.3070  | 0.3080  | 0.2890 |
|           | 2     | 0.3470  | 0.3380  | 0.3470  | 0.3320 |
|           | 3     | 0.3620  | 0.3400  | 0.3620  | 0.3430 |
|           | 4     | 0.3700  | 0.3510  | 0.3700  | 0.3500 |
|           | 5     | 0.3910  | 0.3650  | 0.3910  | 0.3630 |

a substantial decrease in ROUGE-L and BLEU scores, highlighting T5's superior ability to process and handle complex textual data. Additionally, removing cross-attention between image and text embeddings produced incoherent summaries, demonstrating its critical role in maintaining multimodal coherence. These findings confirm that each component of the SumGPT model plays an essential role in generating accurate and contextually rich radiology report summaries.

#### N. QUALITATIVE FEEDBACK AND CLINICAL RELEVANCE ASSESSMENT

A qualitative assessment was conducted with five expert radiologists to evaluate the practical applicability of the proposed SumGPT model. They assessed the generated summaries based on three dimensions: readability, accuracy

TABLE 11. Results of ViLBERT.

| Model   | Epoch | ROUGE-1 | ROUGE-2 | ROUGE-L | BLEU   |
|---------|-------|---------|---------|---------|--------|
| ViLBERT | 1     | 0.4918  | 0.2377  | 0.4918  | 0.4253 |
|         | 2     | 0.5014  | 0.2520  | 0.5014  | 0.4163 |
|         | 3     | 0.4945  | 0.2418  | 0.4945  | 0.4126 |
|         | 4     | 0.5000  | 0.2500  | 0.5000  | 0.4155 |
|         | 5     | 0.5000  | 0.2500  | 0.5000  | 0.4312 |

TABLE 12. Results of T5+BERT.

| Model   | Epoch | ROUGE-1 | ROUGE-2 | ROUGE-L | BLEU   |
|---------|-------|---------|---------|---------|--------|
| T5+BERT | 1     | 0.4973  | 0.2459  | 0.4973  | 0.4145 |
|         | 2     | 0.4993  | 0.2490  | 0.4993  | 0.4226 |
|         | 3     | 0.4959  | 0.2439  | 0.4959  | 0.4268 |
|         | 4     | 0.4980  | 0.2469  | 0.4980  | 0.4235 |
|         | 5     | 0.5068  | 0.2602  | 0.5068  | 0.4345 |

in conveying findings, and usefulness in clinical workflows, using a 5-point scale. The model achieved mean scores of 4.6, 4.4, and 4.7 for these dimensions, respectively. Radiologists found the summaries to be highly readable and concise, providing quick and appropriate references for use on clinical workstations. Crucial diagnostic information was effectively captured, although minor omissions regarding less important incidental findings were noted. Regarding integration with current workflows, the summaries significantly reduced routine case interpretation times and enhanced efficiency. One radiologist commented:

*Summaries are clear, key points are underlined, and this saves time during daily rounds.*

These results validate the use of SumGPT for improving clinical efficiency while maintaining accuracy and readability. The feedback also highlighted areas for refinement, such

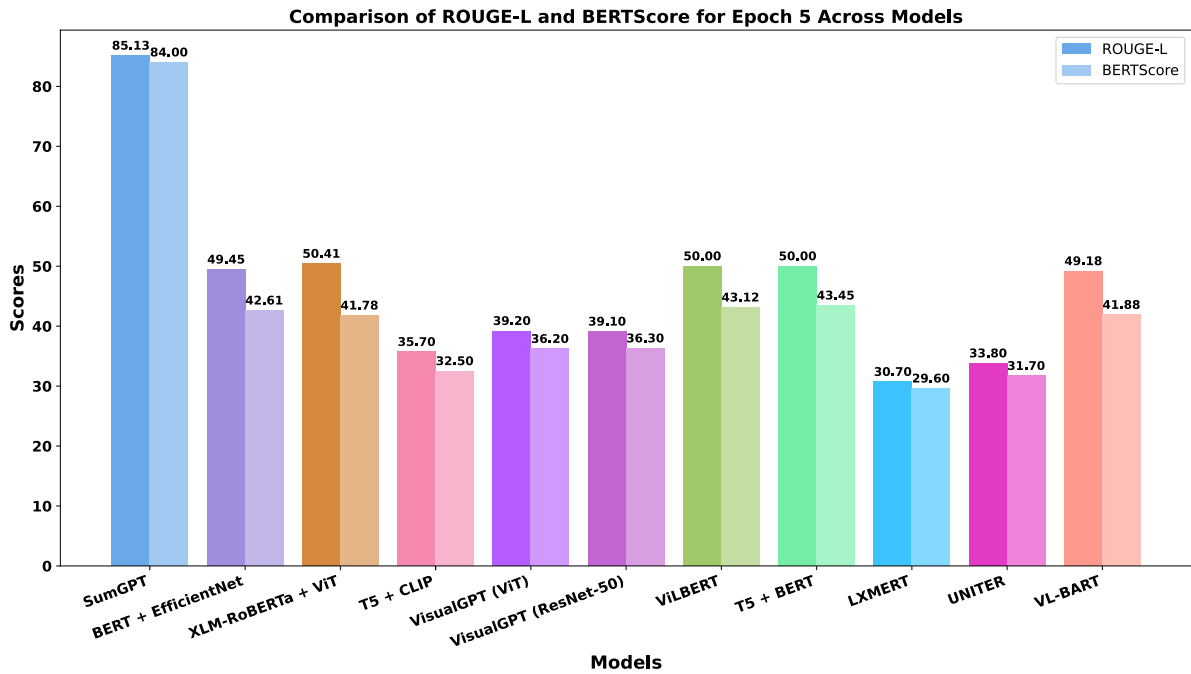


FIGURE 6. Comparison of ROUGE-L and BERTScore for epoch 5 across models.

TABLE 13. Results of LXMERT.

| Model  | Epoch | ROUGE-1 | ROUGE-2 | ROUGE-L | BLEU   |
|--------|-------|---------|---------|---------|--------|
| LXMERT | 1     | 0.2290  | 0.2280  | 0.2290  | 0.2110 |
|        | 2     | 0.2690  | 0.2590  | 0.2690  | 0.2580 |
|        | 3     | 0.2820  | 0.2700  | 0.2820  | 0.2710 |
|        | 4     | 0.2980  | 0.2880  | 0.2980  | 0.2870 |
|        | 5     | 0.3070  | 0.2970  | 0.3070  | 0.2960 |

TABLE 15. Results of VL-BART.

| Model   | Epoch | ROUGE-1 | ROUGE-2 | ROUGE-L | BLEU   |
|---------|-------|---------|---------|---------|--------|
| VL-BART | 1     | 0.4939  | 0.2408  | 0.4939  | 0.4168 |
|         | 2     | 0.5068  | 0.2602  | 0.5068  | 0.4240 |
|         | 3     | 0.5007  | 0.2510  | 0.5007  | 0.4286 |
|         | 4     | 0.4843  | 0.2264  | 0.4843  | 0.4077 |
|         | 5     | 0.4918  | 0.2377  | 0.4918  | 0.4188 |

TABLE 14. Results of Uniter.

| Model  | Epoch | ROUGE-1 | ROUGE-2 | ROUGE-L | BLEU   |
|--------|-------|---------|---------|---------|--------|
| UNITER | 1     | 0.2430  | 0.2420  | 0.2430  | 0.2230 |
|        | 2     | 0.2870  | 0.2800  | 0.2870  | 0.2750 |
|        | 3     | 0.3020  | 0.2910  | 0.3020  | 0.2900 |
|        | 4     | 0.3200  | 0.3080  | 0.3200  | 0.3050 |
|        | 5     | 0.3380  | 0.3260  | 0.3380  | 0.3170 |

as enhancing the coverage of incidental findings, to further improve its clinical utility.

IV. DISCUSSION

The experimental results from evaluating different models provide valuable insights into the effectiveness of various architectures for summarizing radiology reports. This study primarily focused on comparing the performance of the proposed SumGPT model with several other state-of-the-art multimodal models, including BERT + EfficientNet, XLM-RoBERTa + ViT, T5 + CLIP, and VisualGPT (GPT-2 + ViT). The findings highlight the strengths and limitations of

each approach, offering a comprehensive understanding of their capabilities in generating concise and accurate radiology report summaries.

As detailed in Algorithm 1, the cross-modal fusion of T5 and ViT embeddings emerges as a critical factor in achieving superior performance. This fusion mechanism effectively integrates textual and visual data, enabling the model to produce coherent and contextually rich summaries. The results underscore the importance of advanced transformer-based architectures and multimodal learning techniques in addressing the complexities of radiology report summarization.

A. PROPOSED SUMGPT MODEL

The proposed SumGPT model consistently outperformed all other models across multiple evaluation metrics, including ROUGE-1, ROUGE-2, ROUGE-L, and BLEU scores, with peak performance observed at epoch 3. As shown in Table 5, the SumGPT model achieved ROUGE scores exceeding 85% and BLEU scores approaching 85%, underscoring its effectiveness in integrating visual and textual information.

**Algorithm 1** Proposed SumGPT Model

---

```

1: Input: Multimodal Radiology Dataset (Images and Text Reports)
2: Output: Summarized Radiology Reports
3: procedure T5 + ViT Multimodal Model
4:   Initialize: Pre-trained T5 and ViT with weights
5:   Step 1: Split Dataset
6:     a. Training Set (TrS)
7:     b. Testing Set (TeS)
8:   Step 2: Train Model on Training Set
9:      $T5+ViT_{Model} \leftarrow \text{Train}_{T5+ViT}(TrS)$ 
10:  Step 3: Generate Summaries for Testing Set
11:     $Summaries_{Output} \leftarrow \text{Predict}_{T5+ViT}(TeS)$ 
12:  Step 4: Output Summarized Reports
13:  for each image-text pair  $i$  in Testing Set (TeS) do
14:    Generate summary for the given sample:
15:     $Summary_i \leftarrow \text{Predict}_{T5+ViT}(i)$ 
16:  end for
17:    Store all  $Summaries_{Output}$  for further evaluation
18:  Step 5: Evaluate Model Performance
19:    Calculate evaluation metrics (e.g., ROUGE, BLEU):
20:    Metrics  $\leftarrow$ 
21:    Evaluate( $Summaries_{Output}$ , Ground Truth)
22:    if Evaluation is satisfactory then
23:      Conclude that the model generates clinically relevant summaries
24:    else
25:      Adjust parameters and retrain the model
26:    end if
27:  end procedure

```

---

This superior performance can be attributed to the model's robust architecture, which combines the strengths of the T5 transformer for text generation and the ViT for image feature extraction [13]. The results highlight the SumGPT model's ability to capture nuanced details in radiology images and translate them into coherent textual summaries, making it a strong candidate for clinical applications in radiology report generation.

**B. COMPARISON WITH OTHER MODELS**

While the **BERT + EfficientNet** model demonstrated competitive performance, as shown in Table 6, its metrics remained below 80%. This suggests that while BERT + EfficientNet can generate reasonable summaries, it falls short in fully capturing and integrating the complexities of visual and textual data in radiology reports [14], [36]. Similarly, the **XLM-RoBERTa + ViT** model, whose results are detailed in Table 7, performed well but failed to match the SumGPT model's performance. Despite its potential for handling multimodal data, the results indicate that further fine-tuning and optimization are needed to achieve performance comparable to that of SumGPT.

The **T5 + CLIP** and **VisualGPT (GPT-2 + ViT)** models, presented in Table 8 and 9, respectively, also showed strong performance across all metrics. However, their scores consistently remained below 80%. This indicates that while these models are effective, they do not achieve the same accuracy and coherence demonstrated by SumGPT [11]. These findings suggest combining the T5 model with ViT for feature extraction provides a more robust framework for handling the complexities inherent in radiology report summarization.

**C. PERFORMANCE OF OTHER MULTIMODAL MODELS**

Other multimodal models, including **ViLBERT**, **LXMERT**, **UNITER**, and **VL-BART**, as shown in Tables 11, 13, 14, and 15, respectively, demonstrated moderate performance, with all metrics falling below 80%. While these models are theoretically capable of bridging visual and textual modalities, they appeared to struggle with the specific demands of radiology report summarization. This suggests limitations in their ability to fully integrate and interpret the nuanced information in medical images and corresponding textual data. The performance gap between these models and the proposed **SumGPT** model underscores the importance of selecting model architectures that are not only powerful but also well-suited to the specific task and dataset characteristics [14], [15], [36].

**D. IMPLICATIONS OF RADIOLOGY SUMMARIZATION**

The findings from this study underscore the critical importance of integrating visual and textual data for effective radiology report summarization. The superior performance of the **SumGPT** model suggests that models capable of deeply integrating vision and language processing are better equipped to handle the complexities inherent in radiology reports. This has significant implications for clinical practice, where accurate and coherent summarization of radiology reports is essential for effective patient care [41].

The **SumGPT** model represents a substantial opportunity to streamline radiology workflows, particularly in summarizing highly complex reports. Reducing reporting time can alleviate workloads and allow radiologists to focus on interpreting more critical cases. Its compatibility with existing Radiology Information Systems (RIS) and Picture Archiving and Communication Systems (PACS) ensures easy integration, making summaries readily accessible. Additionally, incorporating feedback loops into the system can help ensure alignment with institutional standards. These functionalities demonstrate the practical relevance of **SumGPT** in improving clinical efficiency and patient care outcomes. Furthermore, the results highlight the potential for future advancements by exploring hybrid models that combine the strengths of various architectures. Future work could focus on refining these models through additional fine-tuning and expanding the dataset to include more diverse radiological conditions [41].

**TABLE 16.** Comparison of state-of-the-art models for radiology report summarization.

| Reference             | Model                   | Approach                                                                                                         | Results                                                                     |
|-----------------------|-------------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| [8]                   | METransformer           | A multi-expert transformer model for radiology report generation, incorporating domain-specific expertise.       | ROUGE: 38.0%, BLEU: 17.2%, METEOR: 19.2%                                    |
| [37]                  | Progressive Transformer | A two-step approach that generates global concepts and refines them into coherent text with a transformer model. | IU X-Ray (ROUGE: 39.0%, BLEU: 17.3%), MIMIC-CXR (ROUGE: 27.2%, BLEU: 10.7%) |
| [38]                  | TrMRG                   | ViT and Language Model-Based Radiology Report Generation.                                                        | ROUGE: 40.8%, BLEU: 22.5%                                                   |
| [39]                  | MATNet                  | Exploiting multi-modal features for radiology report generation.                                                 | IU X-Ray (ROUGE: 44.6%, BLEU: 25.4%), MIMIC-CXR (ROUGE: 39.5%, BLEU: 23.3%) |
| [40]                  | XPRONET                 | Cross-modal prototype-driven network for generating radiology reports.                                           | IU X-Ray (ROUGE: 41.1%, BLEU: 19.9%), MIMIC-CXR (ROUGE: 27.9%, BLEU: 20.5%) |
| <b>Proposed Model</b> | <b>SumGPT</b>           | <b>T5 and ViT for integrating text and visual data in radiology reports.</b>                                     | <b>ROUGE-1: 85.14%, ROUGE-2: 84.71%, ROUGE-L: 85.14%, BLEU: 84.70%</b>      |

### E. COMPARING STATE-OF-THE-ART METHODS

Table 16 provides a comprehensive overview of recent advancements in radiology report summarization, summarizing various approaches, methodologies, dataset characteristics, and performance results from notable studies in the field.

For instance, the research in [8] demonstrated the effectiveness of the METransformer model, achieving a ROUGE-L score of 38.0% and a BLEU score of 17.2%. Similarly, the Progressive Transformer method [37] achieved ROUGE and BLEU scores of 39.0% and 17.3%, respectively, on the IU X-Ray dataset. Another notable study [38] employed the TrMRG model, which integrates ViT and language models, and achieved a ROUGE score of 40.8% and a BLEU score of 22.5%. The MATNet model [39] exploited multimodal features, resulting in a ROUGE score of 44.6% and a BLEU score of 25.4% on the IU X-Ray dataset. Similarly, XPRONET [40], leveraging cross-modal prototypes, achieved a ROUGE score of 41.1% and a BLEU score of 19.9% on the IU X-Ray dataset.

Among these state-of-the-art models, the **SumGPT** model consistently demonstrated the highest performance across all metrics, achieving a remarkable ROUGE-1 score of 85.14% and a BLEU score of 84.70% by the third epoch (Table 5). This highlights SumGPT's superior capability to integrate textual and visual modalities, effectively capturing the complex relationships within radiology data to produce accurate and coherent summaries.

In comparison, the **BERT + EfficientNet** (Table 6) and **XLM-RoBERTa + ViT** models (Table 7) showed competitive performance but with lower scores. The BERT + EfficientNet model achieved a maximum ROUGE-1 score of 49.45% and a BLEU score of 36.9% at the fifth epoch, while the XLM-RoBERTa + ViT model reached a ROUGE-1 score of 50.41% and a BLEU score of 41.78%. These results reflect the challenges of integrating transformer architectures with visual data effectively.

The **VisualGPT** models, specifically the combinations of GPT-2 + ViT (Table 9) and GPT-2 + ResNet-50 (Table 10), also performed commendably, with ROUGE-1 scores peaking at 39.2%. However, these models require

further tuning to reach the performance levels achieved by SumGPT. Emerging models such as **ViLBERT** (Table 11), **LXMERT** (Table 13), and **UNITER** (Table 14) showed potential for specific scenarios. For example, ViLBERT achieved a ROUGE-1 score of 50.00%, while UNITER attained a ROUGE-1 score of 33.8%, demonstrating their effectiveness in particular multimodal tasks.

These findings underscore the importance of selecting models based on the specific requirements of the task. While **SumGPT** excels in overall performance, the choice of model may depend on factors such as computational efficiency, ease of integration, and dataset characteristics. Future research directions include targeted fine-tuning of underperforming models like ViLBERT and LXMERT to enhance their applicability to medical tasks [14], [15]. Additionally, exploring few-shot learning could address the challenge of limited data, a common issue in medical imaging applications [42].

### V. CONCLUSION AND FUTURE WORKS

This study demonstrated the effectiveness of integrating visual and textual data for radiology report summarization, primarily using transformer-based models. Among the evaluated models, the proposed **SumGPT** model, which combines the T5 and ViT architectures, consistently outperformed other approaches in generating coherent and accurate summaries. By integrating these two modalities, SumGPT achieved a deeper understanding of the intricate details in radiology reports, significantly enhancing the quality of the generated summaries. The evaluation results, which highlighted superior performance across all primary metrics, underscore the potential of transformer-based models in addressing complex tasks in medical imaging. These models, by effectively capturing visual and textual information, hold great promise for improving efficiency in clinical workflows.

The findings of this study emphasize the value of multimodal approaches in enhancing the interpretability and utility of radiology reports, laying a strong foundation for their application in real-world healthcare environments. Future research should focus on enhancing the generalization

capabilities of these models, particularly in low-resource scenarios. Approaches such as few-shot and zero-shot learning could prove instrumental in achieving robust performance with limited data. Additionally, extending the applicability of these models to other medical domains and exploring their real-time deployment in clinical settings are crucial steps towards maximizing their practical utility and impact in healthcare. While SumGPT performed exceptionally, its dataset's limited diversity in imaging modalities and demographics may affect generalizability. Future work will focus on evaluating SumGPT across varied scenarios to ensure robustness and improve patient care by enhancing multimodal integration and interpretability.

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dashboard creation, power BI, and SQL database management. With more than five years of experience as a Data Analyst and ML researcher across various industries, he brings extensive expertise to his roles. He is proficient in SQL, R, Python, Tableau, and Excel Power BI, he has a strong foundation in ML and a genuine passion for data analysis, making him well-equipped to excel in data science.



journals and conference proceedings, such as IEEE Access, Elsevier, MDPI, Springer, and international conferences. Moreover, he actively engages in partnerships with international researchers, recognizing that research is indispensable in fostering innovation. Overall, he is hardworking and has taught himself various skills, such as NLP, LLM, and DL.

**TIPU SULTAN** received the B.Sc. degree (Hons.) in mechatronics engineering from Kyungsoong University, South Korea, and the M.Sc. degree in data science from Fordham University, Lincoln Center, in April 2024. He is currently pursuing the Ph.D. degree in aerospace and mechanical engineering with Saint Louis University, focusing on applying offline NLP models to robotic arms. He interned as a Data Analyst with the Authentic Brands Group. He specializes in ML, Tableau

**MOHAMMAD ABU TAREQ RONY** received the Bachelor of Science degree in statistics from Noakhali Science and Technology University, Noakhali, Bangladesh. Additionally, he possesses expertise in devising advanced analytics strategies using data. His diverse professional experience includes four years of research in areas, such as artificial intelligence. He is currently a Research Data Scientist with aiQuest Intelligence, Dhaka, Bangladesh. He has published papers in refereed



fields is driven by a quest to uncover hidden insights within data, develop intelligent learning algorithms, bridge the communication gap between humans and machines, and artistically enhance digital imagery. He is a Researcher. His approach is characterized by a blend of technical proficiency and creative problem-solving, aiming to contribute significantly to the frontiers of technology and its application in understanding and improving their digital world.

**MOHAMMAD SHARIFUL ISLAM** received the B.Sc. degree in computer science and telecommunication engineering from Noakhali Science and Technology University, Bangladesh, in 2023, brings a deep passion for cutting-edge technologies to the research community. His academic journey, rooted in computer science and telecommunications, has evolved into a genuine pursuit of specialized areas, including data science, ML, NLP, and image processing. His work in these

**SAMAH ALSHATHRI** received the Bachelor of Computer Science and Master of Computer Engineering degrees from King Saud University, Riyadh, Saudi Arabia, and the Ph.D. degree from the Department of Computer and Mathematics, Plymouth University, Plymouth, U.K. She is currently an Assistant Professor with the Department of Information Technology, College of Computer and Information Sciences, Princess Nourah bint Abdulrahman University (PNU), Riyadh. She was the Chair of the Network and Communication Department and participated in organizing many international conferences. She has authored or co-authored many articles published in well-known journals in the research field. Her research interests include wireless networks, cloud computing, fog computing, the IoT, data mining, ML, text analytics, image classification, and DL.



**WALID EL-SHAFAI** (Senior Member, IEEE) was born in Alexandria, Egypt. He received the B.Sc. degree (Hons.) in electronics and electrical communication engineering from the Faculty of Electronic Engineering (FEE), Menoufia University, Menouf, Egypt, in 2008, the M.Sc. degree from Egypt-Japan University of Science and Technology (E-JUST), in 2012, and the Ph.D. degree from the FEE, Menoufia University, in 2019.

From January 2021 to December 2024, he was a Researcher with the Security Engineering Laboratory (SEL), Prince Sultan University (PSU), Riyadh, Saudi Arabia. He is currently a Senior Researcher with the Automated Systems and Soft Computing Laboratory (ASSCL) and an Assistant Professor with the College of Computer Science and Information Systems, PSU. In addition, he is an Associate Professor with the Department of Electronics and Communication Engineering (ECE), Faculty of Electronic Engineering, Menoufia University. His research interests include a broad range of fields, including wireless mobile and multimedia communication systems, image and video signal processing, efficient 2D/3D video coding and transmission, quality of service and experience, digital communication techniques, cognitive radio networks, adaptive filter design, 3D video watermarking, steganography, encryption, error resilience and concealment algorithms for video codecs (H.264/AVC, H.264/MVC, and H.265/HEVC), cognitive cryptography, medical image processing, speech processing, security algorithms, software-defined networks, the Internet of Things, FPGA implementations of signal processing algorithms and communication systems, cancellable biometrics, pattern recognition, image and video magnification, artificial intelligence applications in signal processing and communication systems, modulation identification and classification, image and video super-resolution and denoising, automated systems, cybersecurity applications, malware and ransomware detection, and DL for signal processing and communication systems. He is a dedicated reviewer for numerous international journals and conferences, contributing to the advancement of research in his areas of expertise.

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